数理分析方法

集合论与集合运算

课件编号：102-ZSM-PPT　原始 PPT：53 页

逐页保留原始 PPT 画面 + 中文零基础自学讲解 + 证明专题

阅读说明：本讲义每一页原课件均以原始 PDF 页面完整嵌入；其后的中文解读旨在帮助初学者识别定义、假设、论证步骤。章末为从零展开的核心证明与复习提示。讲义属于学习辅助材料，不代替原教师课件。参考：Gérard Debreu (1959), *Theory of Value*。

目录

1 学前导读

这一讲学什么

集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。
先读“属于”的条件，再化成逻辑上的“且、或、非”。特别分清元素 x A 与子集 B A。

证明题通用方法（先读本页）

严格证明的四个必要动作

第一：写出假设。比如“K 非空、闭、有界”，不能只写“K 很好”。
第二：逐字展开定义。写出 $\forall$（任意）、$\exists$（存在）、$\Rightarrow$（推出）的顺序。
第三：解释每一次定理调用。从有界得到收敛子列还缺 Bolzano–Weierstrass；得到极限在原集合还需要闭性。
第四：检查结论是否恰好是题目所求。“有极限”不等于“极限属于集合”，“函数有界”不等于“达到最大值”。

最常用的三种模板

集合相等：任取 x，证明 $x \in A \Rightarrow x \in B$，再证明 $x \in B \Rightarrow x \in A$。
极限：给定任意 $\varepsilon > 0$，找出一个 N（或 δ）并验证最终不等式。
存在性：常用反证、上确界构造、嵌套区间、连续函数极值、固定点定理；对每个方法要明确其全部前提。

2 逐页课件与中文讲解

2.1 第 1 页｜本页课件

Theory of Value

An Axiomatic Analysis of Economic Equilibrium

Gerard Debreu (1959)

Professor Shunming Zhang

School of Finance
Renmin University of China

September 10, 2022

这一页怎么读

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2.2 第 2 页｜ Chapter 1: Mathematics

Chapter 1: Mathematics

01. Introduction
02. Sets
03. Functions and Correspondences
04. Preorderings
05. Real Numbers
06. Limits in $\mathbb{R}^N$
07. Continuous Functions
08. Continuous Correspondences
09. Vectors in $\mathbb{R}^N$
10. Fixed Points

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本页重点：反身性是 xRx；传递性是 xRy 且 yRz　xRz。完全性与反对称性不包含在预序定义内。

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2.3 第 3 页 | Sets

Sets

Set Theory is created by **Georg Cantor**[1] in 1880s, and has been developed into one of independent branches in mathematics. Its basic concepts and methods have been infiltrated into various fields of mathematics in the 20th century and become a feature of modern mathematics. **Set Theory** is to explore various properties of sets, its original works is closely related to the in-depth study of mathematical analysis, and lays a foundation for the development of real analysis. This section only gives a necessary introduction to general sets and point-set knowledge in $\mathbb{R}^N$.

[1] Georg Ferdinand Ludwig Philipp Cantor (March 3, 1845 - January 3, 1918), German mathematician and founder of set theory.

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2.4 第 4 页｜本页课件

Sets

In the middle school learning stage, we have come into contact with some preliminary knowledge about sets, such as all natural numbers form a set, often written as $\mathbb{N}$; all rational numbers form a set, often written as $\mathbb{Q}$; all the real numbers form a set, often called $\mathbb{R}$; and so on. So what exactly is a set/collection? Set is the most primitive concept in mathematics, which can not be defined precisely. According to Cantor's original statement (which can be called generality), set refers to all objects that have certain characteristics or meet certain properties as a whole, this whole is called a set, and these objects are called elements of the set. Therefore, the so-called given set or the existence of a set means that a certain constraint has been determined, and it can determine whether any object belongs to the set (in other words, the relationship between an object and a given set has only the relationship that belongs to or does not belong to, and nothing else.). This descriptive definition is sufficient for our practical application.

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2.5 第 5 页 | Sets

Sets

Why does one introduce the concept of set? The first is to examine the overall characteristics and structure of a certain object, study the abstract commonality after discarding the individuality of objects, and divide the sphere of influence. The second is that the language of set theory is very concise, and can distinguish the various connotations of the research objects in more detail. When examining the possibility of their various combinations, it is also very general. This viewpoint and method have appeared as early as in the mathematical work in the 18th century, for example, studying all curves passing through a point on a surface, all possible motions in a mechanical system, and so on. However, the real impetus to the systematic study of set theory comes from the exploration of real numbers set structure caused by the rigor of analysis. For example, **Georg Cantor** has discussed the classification of discontinuous points of functions.

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2.6 第 6 页｜ Definition

Definition

A set S is a collection of objects of any kind (for example points in a plane, real numbers, functions) which are called the elements (or points) of S; sometimes one says a class (system, family) of elements.

Example: (1) The set A of actions available to a certain social agent and among which he has to choose. (2) The set $\mathbb{N}$ of positive integers, or whole numbers, $\{1, 2, \cdots\}$ (braces will denote sets).

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2.7 第 7 页｜ Definition

Definition

The sets which constitute the universe of discourse must always be explicitly listed at the outset.

Example: A social system may be described as composed of a certain number I of agents. Any one of these agents may be indicated by a positive integer n, one of the positive integers $1, \cdots, I$. For any one of them, say the ith one, a set A_i of available actions is given. The universe of discourse consists of the I sets $A_1, \cdots, A_I$.

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2.8 第 8 页｜ Definition

Definition

$x \in S$ $[S \ni x]$ expresses that x denotes a certain element of S; it is read: x belongs to S, or x is an element of S, or x is in S, or S owns x.

Example: $a_i \in A_i$, indicates that a_i is an action available to the ith agent.

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2.9 第 9 页｜ Definition

Definition

If x and x' denote elements of S, then $x = x'$ (x equals x') expresses that they denote the same element, and $x \neq x'$, (x different from x') expresses that they denote different elements.

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2.10 第 10 页｜Definition

Definition

Let $\mathscr{P}$ be a property which any element x of S has or does not have. $\{x \in S \mid x \text{ has property } \mathscr{P}\}$ denotes the set of all the elements of S which have property $\mathscr{P}$; it may be read: the set of x in S such that x has property $\mathscr{P}$.

Example: If A is the set of actions available to an agent and a' is an element of A, the set of those available actions which that agent considers more desirable than a' is $\{a \in A \mid a \text{ is preferred to } a'\}$.

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2.11 第 11 页｜ Subsets

Subsets

A set X of elements of S is called a subset of S; this is denoted by $X \subseteq S$ (X is contained in S). Or equivalently, for two sets X and S, if every member of set X is also a member of set S,

$$x \in X \qquad \Longrightarrow \qquad x \in S,$$

we write $X \subseteq S$ and we also say that X is a subset of S or that S contains X and write $S \supseteq X$. If $X \subseteq S$ and S is not contained in X ($S \nsubseteq X$), we say that X is a proper subset of S and write $X \subset S$ (X is strictly contained in S). If $X \subseteq S$ and $S \subseteq X$, we write $X = S$ (X is equal to S). In this case $x \in X$ is equivalent to $x \in S$.

Example to emphasize the last point: $\mathbb{N} \subseteq \mathbb{N}$ (the set of positive integers).

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2.12 第 12 页｜ Subsets

Subsets

We write $x \notin S$ if it is not true that $x \in S$; $X \not\supseteq S$ if it is not true that $X \supseteq S$; $S \neq X$ if it is not true that $S = X$, etc. A subset of S is said to be empty if it contains no elements; this set is denoted by $\varnothing$. A subset of S is said to be full if it is equal to the set S. The set consisting of the elements, $x_1, \cdots, x_N$, is denoted by $\{x_1, \cdots, x_N\}$. More generally, the set of elements of S which satisfy a property $\mathscr{P}$ is denoted by

$$\{x \in S \mid x \text{ satisfies } \mathscr{P}\}.$$

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读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.13 第 13 页｜ Subsets

Subsets

Careful distinction must be made between the element x of S and the subset $\{x\}$ of S having the only element x.

Example: The assertion $x \in X$ is equivalent to the assertion $\{x\} \subseteq X$.

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2.14 第 14 页｜ Subsets

Subsets

A property $\mathscr{P}$ defines a subset of S, namely the set of elements of S having that property. When no element of S has the property $\mathscr{P}$, one says that $\mathscr{P}$ defines the empty subset of S denoted $\varnothing$. This convention is necessary if a property is always to define a subset of S.

Example: In the example of 1.2.e. $\{a \in A \mid a \text{ is preferred to } a'\}$ is always a subset of A. If a' happens to be the most desirable action in A, then the above set is the empty subset of A.

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2.15 第 15 页｜Subsets

Subsets

Let Λ be a given set. For each $\lambda \in \Lambda$, we specify a set A_λ. In this way, we get many sets, whose total is called the set family, which is written as $\{A_\lambda \mid \lambda \in \Lambda\}$ or $\{A_\lambda\}_{\lambda\in\Lambda}$. Here, Λ is often referred to as an indicator set. When $\Lambda = \mathbb{N}$, a set family is also called a set sequence, which is abbreviated as $\{A_n; n \in \mathbb{N}\}$, and so on.

Example: Let x, y, z be three different numbers, and $A_1 = \{x, y, z\}$, $A_2 = \{x^2, y^2, z^2\}$, $A_3 = \{xy, yz, zx\}$. If $A_1 = A_2 = A_3$, then $\{x, y, z\} = \{1, w, w^2\}$, where $w = \dfrac{-1 \pm \sqrt{3}i}{2}$, i is an imaginary unit.

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2.16 第 16 页｜Basic Operation: Unions and Intersections

Basic Operation: Unions and Intersections

The decomposition and composition of sets is an effective means to explore the relationship between sets and to form new sets such that the set theory method has an important application in the theory of real analysis. This kind of decomposition and synthesis can be expressed by the operation between various sets. Now, its concepts and main properties are briefly introduced.

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2.17 第 17 页｜本页课件

Basic Operation: Unions

Two sets can be "added" together. The union of A_1 and A_2, denoted by $A_1 \cup A_2$, is the set of all things that are members of either A_1 or A_2 (the set of all objects which belong to A_1 or to A_2, also includes those objects belonging to both A_1 and A_2).

$$A_1 \cup A_2 = \{x; x \in A_1 \text{ or } x \in A_2 \text{ or both}\}.$$

Note the common members are accounted only once.

Some basic properties of unions are listed as follows.

(1) $A_1 \cup A_2 = A_2 \cup A_1$ (Abelian or Commutativity);
(2) $A_1 \cup (A_2 \cup A_3) = (A_1 \cup A_2) \cup A_3$ (Associativity);
(3) $A_1 \subseteq A_1 \cup A_2$;
(4) $A \cup A = A$;
(5) $A \cup S = S$;
(6) $A \cup \varnothing = A$;
(7) $A_1 \subseteq A_2$ if and only if $A_1 \cup A_2 = A_2$.

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2.18 第 18 页｜Basic Operation: Unions

Basic Operation: Unions

We extend the union of two sets to general case of multiple sets. Let $\mathscr{X}$ be a family of subsets of S. The union of these subsets, $\bigcup_{X\in\mathscr{X}} X$, is the set of elements of S which belong to at least one X in $\mathscr{X}$.

For a set Λ, suppose that for each $\lambda \in \Lambda$, there is defined a set X_λ. Let $\mathscr{X}$ be the collection of sets X_λ for $\lambda \in \Lambda$. We write $\mathscr{X} = \{X_\lambda | \lambda \in \Lambda\}$ and refer to this as an indexing (or parametrization) of $\mathscr{X}$ by the index set (or parameter set) Λ. The union of all sets in $\mathscr{X}$ is defined by

$$\begin{aligned}\bigcup_{\lambda\in\Lambda} X_\lambda &= \{x : x \in X_\lambda \text{ for at least one } \lambda \in \Lambda\} \\ &= \{x : \text{there exists } \lambda \in \Lambda \text{ such that } x \in X_\lambda\}.\end{aligned}$$

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2.19 第 19 页｜Basic Operation: Unions

Basic Operation: Unions

In particular, when $\Lambda = \{1, \cdots, N\} = \mathcal{N}$, then

$$\begin{aligned} \bigcup_{n=1}^{N} A_n &= \{x : x \in A_n \text{ for at least one } n \in \mathcal{N}\} \\ &= \{x : \text{there exists } n \in \mathcal{N} \text{ such that } x \in A_n\}; \end{aligned}$$

and when $\Lambda = \{1, 2, \cdots\} = \mathbb{N}$, then

$$\begin{aligned} \bigcup_{n=1}^{\infty} A_n &= \{x : x \in A_n \text{ for at least one } n \in \mathbb{N}\} \\ &= \{x : \text{there exists } n \in \mathbb{N} \text{ such that } x \in A_n\}. \end{aligned}$$

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2.20 第 20 页｜ Basic Operation: UnionsExample: If An = {n} for n = 1,

Basic Operation: Unions

Example: If $A_n = \{n\}$ for $n = 1, 2, \cdots$, then

$$\bigcup_{n=1}^{N} A_n = \{1, \cdots, N\} = \mathcal{N} \quad \text{and} \quad \bigcup_{n=1}^{\infty} A_n = \{1, 2, \cdots\} = \mathbb{N}.$$

Example: If $A_n = \{x : n - 1 < x \leqslant n\} = (n - 1, n]$ for $n = 1, 2, \cdots$, then

$$\bigcup_{n=1}^{N} A_n = \{x : 0 < x \leqslant N\} = (0, N] \quad \text{and} \quad \bigcup_{n=1}^{\infty} A_n = \{x : 0 < x < \infty\} = (0, \infty).$$

Example: If $A_n = \left\{x : -1 + \frac{1}{n} \leqslant x \leqslant 1 - \frac{1}{n}\right\} = \left[-1 + \frac{1}{n}, 1 - \frac{1}{n}\right]$ for $n = 1, 2, \cdots$, then

$$\bigcup_{n=1}^{N} A_n = \left\{x : -1 + \frac{1}{N} \leqslant x \leqslant 1 - \frac{1}{N}\right\} = \left[-1 + \frac{1}{N}, 1 - \frac{1}{N}\right] = A_N$$

$$\bigcup_{n=1}^{\infty} A_n = \{x : -1 < x < 1\} = (-1, 1).$$

Example: If $\Lambda = \{\lambda \in \mathbb{Q} \mid 0 < \lambda < 1\}$, and $A_\lambda = \left\{x : \frac{\lambda}{2} < x < 2\right\}$ for $\lambda \in \Lambda$, then

$$\bigcup_{\lambda \in \Lambda} A_\lambda = (0, 2).$$

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

结合原文理解：先读“属于”的条件，再化成逻辑上的“且、或、非”。特别分清元素 x A 与子集 B A。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.21 第 21 页｜本页课件

Basic Operation: Intersections

A new set can also be constructed by determining which members two sets have "in common". The intersection of A_1 and A_2, denoted by $A_1 \cap A_2$, is the set of all things that are members of both A_1 and A_2.

$$A_1 \cap A_2 = \{x : x \in A_1 \text{ and } x \in A_2\}.$$

Some basic properties of intersections are listed as follows.

(1) $A_1 \cap A_2 = A_2 \cap A_1$ (Abelian or Commutativity);
(2) $A_1 \cap (A_2 \cap A_3) = (A_1 \cap A_2) \cap A_3$ (Associativity);
(3) $A_1 \cap A_2 \subseteq A_1$;
(4) $A \cap A = A$;
(5) $A \cap S = A$;
(6) $A \cap \varnothing = \varnothing$;
(7) $A_1 \subseteq A_2$ if and only if $A_1 \cap A_2 = A_1$.

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

结合原文理解：先读"属于"的条件，再化成逻辑上的"且、或、非"。特别分清元素 x A 与子集 B A。

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2.22 第 22 页｜Basic Operation: Intersections

Basic Operation: Intersections

We extend the intersection of two sets to general case of multiple sets. Let $\mathscr{X}$ be a family of subsets of S. The intersection of these subsets, $\bigcap_{X\in\mathscr{X}} X$, is the set of elements of S which belong to all sets X in $\mathscr{X}$.

For a set Λ, suppose that for each $\lambda \in \Lambda$, there is defined a set X_λ. Let $\mathscr{X}$ be the collection of sets X_λ for $\lambda \in \Lambda$. We write $\mathscr{X} = \{X_\lambda | \lambda \in \Lambda\}$ and refer to this as an indexing (or parametrization) of $\mathscr{X}$ by the index set (or parameter set) Λ. The intersection of all sets in $\mathscr{X}$ is defined by

$$\bigcap_{\lambda\in\Lambda} X_\lambda = \{x : x \in X_\lambda \text{ for every } \lambda \in \Lambda\}.$$

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

结合原文理解：先读“属于”的条件，再化成逻辑上的“且、或、非”。特别分清元素 x A 与子集 B A。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.23 第 23 页｜Basic Operation: Intersections

Basic Operation: Intersections

In particular, when $\Lambda = \{1, \cdots, N\} = \mathcal{N}$, then

$$\bigcap_{n=1}^{N} A_n = \{x : x \in A_n \text{ for every } n \in \mathcal{N}\};$$

and when $\Lambda = \{1, 2, \cdots\} = \mathbb{N}$, then

$$\bigcap_{n=1}^{\infty} A_n = \{x : x \in A_n \text{ for every } n \in \mathbb{N}\}.$$

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

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2.24 第 24 页｜ Example: If An =x : 0 ≤x < 1 + 1

Basic Operation: Intersections

Example: If $A_n = \left\{x : 0 \leqslant x < 1 + \frac{1}{n}\right\} = \left[0, 1 + \frac{1}{n}\right)$ for $n = 1, 2, \cdots$, then

$$\bigcap_{n=1}^{N} A_n = \left\{x : 0 \leqslant x < 1 + \frac{1}{N}\right\} = \left[0, 1 + \frac{1}{N}\right) = A_N \text{ and } \bigcap_{n=1}^{\infty} A_n = \{x : 0 \leqslant x \leqslant 1\} = [0, 1].$$

Example: If $A_n = \left\{x : n \leqslant x \leqslant n + \frac{3}{2}\right\}$ for $n = 1, 2, \cdots$, then

$$\bigcap_{n=1}^{3} A_n = \varnothing \quad \text{and} \quad \bigcap_{n=1}^{\infty} A_n = \varnothing.$$

Example: If $A_n = \left\{x : -\frac{1}{n} < x < \frac{1}{n}\right\} = \left(-\frac{1}{n}, \frac{1}{n}\right)$ for $n = 1, 2, \cdots$, then

$$\bigcap_{n=1}^{N} A_n = \left\{x : -\frac{1}{N} < x < \frac{1}{N}\right\} = \left(-\frac{1}{N}, \frac{1}{N}\right) = A_N \quad \text{and} \quad \bigcap_{n=1}^{\infty} A_n = \{0\}.$$

Example: If $\Lambda = \mathbb{R}$, and $A_\lambda = \{x : \lambda \leqslant x < \infty\}$ for $\lambda \in \Lambda$, then

$$\bigcap_{\lambda \in \Lambda} A_\lambda = \varnothing.$$

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

结合原文理解：先读“属于”的条件，再化成逻辑上的“且、或、非”。特别分清元素 x A 与子集 B A。

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2.25 第 25 页｜本页课件

Basic Operation: Unions and Intersections

If A_1 and A_2 have no element in common, i.e., $A_1 \cap A_2 = \varnothing$, then A_1 and A_2 are said to be disjoint. If A_1 and A_2 are two disjoint sets ($A_1 \cap A_2 = \varnothing$), then $A_1 \cup A_2 = A_1 + A_2$.

The sets belonging to a family of sets $\mathscr{X}$ are pairwise disjoint if any two distinctly indexed sets in $\mathscr{X}$ are disjoint. Let $\{X_\lambda | \lambda \in \Lambda\}$ be the collection of sets. If, for any $\lambda_1 \in \Lambda$ and $\lambda_2 \in \Lambda$ with $\lambda_1 \neq \lambda_2$, $X_{\lambda_1} \cap X_{\lambda_2} = \varnothing$, then this collection of sets is pairwise disjoint.

If $\{A_\lambda | \lambda \in \Lambda\}$ is the collection of pairwise disjoint sets, we write the union as addition.

$$\bigcup_{\lambda \in \Lambda} A_\lambda = \sum_{\lambda \in \Lambda} A_\lambda.$$

In particular, when $\{A_n : n = 1, \cdots, N\}$ is a sequence of pairwise disjoint sets, then

$$\bigcup_{n=1}^{N} A_n = \sum_{n=1}^{N} A_n;$$

and when $\{A_n : n = 1, 2, \cdots\}$ is a sequence of pairwise disjoint sets, then

$$\bigcup_{n=1}^{\infty} A_n = \sum_{n=1}^{\infty} A_n.$$

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

结合原文理解：先读“属于”的条件，再化成逻辑上的“且、或、非”。特别分清元素 x A 与子集 B A。

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2.26 第 26 页｜Basic Operation: Unions and Intersections

Basic Operation: Unions and Intersections

Theorem 1. Let A_1, A_2 and A_3 be sets.

(1) Abelian or Commutativity:
- (1.1) $A_1 \cup A_2 = A_2 \cup A_1$;
- (1.2) $A_1 \cap A_2 = A_2 \cap A_1$.

(2) Associativity:
- (2.1) $A_1 \cup (A_2 \cup A_3) = (A_1 \cup A_2) \cup A_3$;
- (2.2) $A_1 \cap (A_2 \cap A_3) = (A_1 \cap A_2) \cap A_3$.

(3) Distributivity:
- (3.1) $A_1 \cup (A_2 \cap A_3) = (A_1 \cup A_2) \cap (A_1 \cup A_3)$;
- (3.2) $A_1 \cap (A_2 \cup A_3) = (A_1 \cap A_2) \cup (A_1 \cap A_3)$.

(4) Absorption:
- (4.1) $A \cup (A \cap A') = A$;
- (4.2) $A \cap (A \cup A') = A$.

(5) Idempotence:
- (5.1) $A \cup A = A$;
- (5.2) $A \cap A = A$.

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

结合原文理解：先读“属于”的条件，再化成逻辑上的“且、或、非”。特别分清元素 x A 与子集 B A。

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2.27 第 27 页｜Basic Operation: Unions and IntersectionsTheorem 2.

Basic Operation: Unions and Intersections

Theorem 2.

(1) If $A_\lambda \subseteq A'_\lambda$ for $\lambda \in \Lambda$, then $\bigcup_{\lambda\in\Lambda} A_\lambda \subseteq \bigcup_{\lambda\in\Lambda} A'_\lambda$. In particular, if $A_\lambda \subseteq A'$ for $\lambda \in \Lambda$, then $\bigcup_{\lambda\in\Lambda} A_\lambda \subseteq A'$.

(2) If $A_\lambda \subseteq A'_\lambda$ for $\lambda \in \Lambda$, then $\bigcap_{\lambda\in\Lambda} A_\lambda \subseteq \bigcap_{\lambda\in\Lambda} A'_\lambda$. In particular, if $A \subseteq A'_\lambda$ for $\lambda \in \Lambda$, then $A \subseteq \bigcap_{\lambda\in\Lambda} A'_\lambda$.

(3) $\bigcup_{\lambda\in\Lambda} (A_\lambda \cup A'_\lambda) = \left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) \cup \left(\bigcup_{\lambda\in\Lambda} A'_\lambda\right)$.

(4) $\bigcap_{\lambda\in\Lambda} (A_\lambda \cap A'_\lambda) = \left(\bigcap_{\lambda\in\Lambda} A_\lambda\right) \cap \left(\bigcap_{\lambda\in\Lambda} A'_\lambda\right)$.

(5) $A \cap \left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) = \bigcup_{\lambda\in\Lambda} (A \cap A_\lambda)$.

(6) $A \cup \left(\bigcap_{\lambda\in\Lambda} A_\lambda\right) = \bigcap_{\lambda\in\Lambda} (A \cup A_\lambda)$.

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

结合原文理解：先读“属于”的条件，再化成逻辑上的“且、或、非”。特别分清元素 x A 与子集 B A。

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2.28 第 28 页｜Basic Operation: Complements

Basic Operation: Complements

Two sets can also be "subtracted". The relative complement of A_2 in A_1 (also called the set-theoretic difference of A_1 and A_2), denoted by $A_1 \setminus A_2$, is the set of all elements that are members of A_1 but not members of A_2.

$$A_1 \setminus A_2 = \{x : x \in A_1 \text{ and } x \notin A_2\}.$$

Note that it is valid to "subtract" members of a set that are not in the set. Generally, $(A_1 \setminus A_2) \cup A_2$ must not be equal to A_1. We can check $(A_1 \setminus A_2) \cup A_2 = A_1 \cup A_2$.

In certain settings all sets under discussion are considered to be subsets of a given universal set S. If X is a subset of S, and x an element of S, then $x \notin X$ expresses that x is not an element of X (x does not belong to X). The elements of S which do not belong to X form a set called the complement of X in S (the absolute complement or simply complement of X relative to S) and denoted by $S \setminus X = \mathbf{C}_S X$. When there can be no ambiguity about S one says the complement of X, and one writes $\mathbf{C}X$.

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

结合原文理解：先读“属于”的条件，再化成逻辑上的“且、或、非”。特别分清元素 x A 与子集 B A。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.29 第 29 页｜Basic Operation: Complements

Basic Operation: Complements

Theorem 3.

(1) $\mathbf{C}S = \varnothing$ and $\mathbf{C}\varnothing = S$;

(2) $A \cup (\mathbf{C}A) = S$ and $A \cap (\mathbf{C}A) = \varnothing$;

(3) $\mathbf{C}(\mathbf{C}A) = A$;

(4) If $A_1 \supseteq A_2$, then $\mathbf{C}A_1 \subseteq \mathbf{C}A_2$;

(5) If $A_1 \cap A_2 = \varnothing$, then $A_1 \subseteq \mathbf{C}A_2$ and $A_2 \subseteq \mathbf{C}A_1$.

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

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2.30 第 30 页｜Basic Operation: Complements

Basic Operation: Complements

Theorem 4. De Morgan's Law.

(1) $\mathbf{C}(A_1 \cup A_2) = \mathbf{C}A_1 \cap \mathbf{C}A_2$ and $\mathbf{C}(A_1 \cap A_2) = \mathbf{C}A_1 \cup \mathbf{C}A_2$.

(2) $\mathbf{C}\left(\bigcup_{n=1}^{N} A_n\right) = \bigcap_{n=1}^{N} \mathbf{C}A_n$ and $\mathbf{C}\left(\bigcap_{n=1}^{N} A_n\right) = \bigcup_{n=1}^{N} \mathbf{C}A_n$.

(3) $\mathbf{C}\left(\bigcup_{n=1}^{\infty} A_n\right) = \bigcap_{n=1}^{\infty} \mathbf{C}A_n$ and $\mathbf{C}\left(\bigcap_{n=1}^{\infty} A_n\right) = \bigcup_{n=1}^{\infty} \mathbf{C}A_n$.

(4) $\mathbf{C}\left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) = \bigcap_{\lambda\in\Lambda} \mathbf{C}A_\lambda$ and $\mathbf{C}\left(\bigcap_{\lambda\in\Lambda} A_\lambda\right) = \bigcup_{\lambda\in\Lambda} \mathbf{C}A_\lambda$.

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

本页重点：德摩根律把补集与交并互换：否定“或”得到“且非”，否定“且”得到“或非”。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「demorgan」证明专题。

2.31 第 31 页｜本页课件

Basic Operation: Symmetric Difference

In mathematics, the symmetric difference, also known as the disjunctive union, of two sets is the set of elements which are in either of the sets and not in their intersection. The symmetric difference of the sets A_1 and A_2 is commonly denoted by $A_1 \triangle A_2$. The symmetric difference is equivalent to the union of both relative complements,

$$A_1 \triangle A_2 = (A_1 \setminus A_2) \cup (A_2 \setminus A_1).$$

The symmetric difference can also be expressed as the union of the two sets, minus their intersection,

$$A_1 \triangle A_2 = (A_1 \cup A_2) \setminus (A_1 \cap A_2).$$

In particular, $A_1 \triangle A_2 \subseteq A_1 \cup A_2$; the equality in this non-strict inclusion occurs if and only if A_1 and A_2 are disjoint sets ($A_1 \triangle A_2 = A_1 \cup A_2$ if and only if $A_1 \cap A_2 = \varnothing$). Furthermore, $A_1 \triangle A_2$ and $A_1 \cap A_2$ are always disjoint, so $A_1 \triangle A_2$ and $A_1 \cap A_2$ partition $A_1 \cup A_2$.

$$A_1 \cup A_2 = (A_1 \triangle A_2) + (A_1 \cap A_2).$$

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

本页重点：对称差 A B 恰是“属于 A 或 B，但不同时属于两者”的元素；可用真值表逐格验证。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

对应的完整证明：参见本讲义最后的「symdiff」证明专题。

2.32 第 32 页｜Basic Operation: Symmetric Difference

Basic Operation: Symmetric Difference

The symmetric difference is commutative and associative (and consequently the leftmost set of parentheses in the previous expression were thus redundant):

(1) Commutativity: $A_1 \triangle A_2 = A_2 \triangle A_1$;

(2) Associativity: $(A_1 \triangle A_2) \triangle A_3 = A_1 \triangle (A_2 \triangle A_3)$.

The empty set is neutral, and every set is its own inverse:

(1) $A \triangle \varnothing = A$;

(2) $A \triangle A = \varnothing$;

(3) $A \triangle \mathbf{C}A = S$;

(4) $A \triangle S = \mathbf{C}A$.

$$(A_1 \triangle A_2) \triangle A_2 = A_1 \triangle (A_2 \triangle A_2) = A_1 \triangle \varnothing = A_1$$

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

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对应的完整证明：参见本讲义最后的「symdiff」证明专题。

2.33 第 33 页｜Basic Operation: Symmetric Difference

Basic Operation: Symmetric Difference

$$
\begin{aligned}
& (A_1 \triangle A_2) \triangle A_3 = [(A_1 \setminus A_2) \cup (A_2 \setminus A_1)] \triangle A_3 \\
= \; & \{[(A_1 \setminus A_2) \cup (A_2 \setminus A_1)] \setminus A_3\} \cup \{A_3 \setminus [(A_1 \cup A_2) \setminus (A_1 \cap A_2)]\} \\
= \; & \{[(A_1 \setminus A_2) \setminus A_3] \cup [(A_2 \setminus A_1) \setminus A_3]\} \cup \{A_3 \cap [\mathbf{C}(A_1 \cup A_2) \cup (A_1 \cap A_2)]\} \\
= \; & [A_1 \cap \mathbf{C}(A_2 \cup A_3)] \cup [A_2 \cap \mathbf{C}(A_3 \cup A_1)] \cup [A_3 \cap \mathbf{C}(A_1 \cup A_2)] \cup [A_3 \cap (A_1 \cap A_2)] \\
= \; & [A_1 \setminus (A_2 \cup A_3)] \cup [A_2 \setminus (A_3 \cup A_1)] \cup [A_3 \setminus (A_1 \cup A_2)] \cup [A_1 \cap A_2 \cap A_3]
\end{aligned}
$$

$$
\begin{aligned}
& A_1 \triangle (A_2 \triangle A_3) = A_1 \triangle [(A_2 \setminus A_3) \cup (A_3 \setminus A_2)] \\
= \; & \{A_1 \setminus [(A_2 \cup A_3) \setminus (A_2 \cap A_3)]\} \cup \{[(A_2 \setminus A_3) \cup (A_3 \setminus A_2)] \setminus A_1\} \\
= \; & \{A_1 \cap [\mathbf{C}(A_2 \cup A_3) \cup (A_2 \cap A_3)]\} \cup \{[(A_2 \setminus A_3) \setminus A_1] \cup [(A_3 \setminus A_2) \setminus A_1]\} \\
= \; & [A_1 \cap \mathbf{C}(A_2 \cup A_3)] \cup [A_1 \cap (A_2 \cap A_3)] \cup [A_2 \cap \mathbf{C}(A_3 \cup A_1)] \cup [A_3 \cap (A_1 \cap A_2)] \\
= \; & [A_1 \setminus (A_2 \cup A_3)] \cup [A_1 \cap A_2 \cap A_3] \cup [A_2 \setminus (A_3 \cup A_1)] \cup [A_3 \setminus (A_1 \cup A_2)] \\
= \; & [A_1 \setminus (A_2 \cup A_3)] \cup [A_2 \setminus (A_3 \cup A_1)] \cup [A_3 \setminus (A_1 \cup A_2)] \cup [A_1 \cap A_2 \cap A_3]
\end{aligned}
$$

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

本页重点：对称差 A　B 恰是“属于 A 或 B，但不同时属于两者”的元素；可用真值表逐格验证。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论?

对应的完整证明：参见本讲义最后的「symdiff」证明专题。

2.34 第 34 页｜Basic Operation: Symmetric Difference

Basic Operation: Symmetric Difference

From the property of the inverses in a Boolean group, it follows that the symmetric difference of two repeated symmetric differences is equivalent to the repeated symmetric difference of the join of the two multisets, where for each double set both can be removed. In particular:

$$(A_1 \triangle A_2) \triangle (A_2 \triangle A_3) = A_1 \triangle A_3.$$

This implies triangle inequality: the symmetric difference of A_1 and A_3 is contained in the union of the symmetric difference of A_1 and A_2 and that of A_2 and A_3.

$$(A_1 \triangle A_2) \triangle (A_2 \triangle A_3) = [(A_1 \triangle A_2) \triangle A_2] \triangle A_3 = A_1 \triangle A_3$$
$$(A_1 \triangle A_2) \triangle (A_2 \triangle A_3) = A_1 \triangle [A_2 \triangle (A_2 \triangle A_3)] = A_1 \triangle A_3$$

这一页怎么读

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2.35 第 35 页｜Basic Operation: Symmetric Difference

Basic Operation: Symmetric Difference

Intersection distributes over symmetric difference:

$$A_1 \cap (A_2 \triangle A_3) = (A_1 \cap A_2) \triangle (A_1 \cap A_3).$$

and this shows that the power set of S becomes a ring with symmetric difference as addition and intersection as multiplication. This is the prototypical example of a Boolean ring. Further properties of the symmetric difference:

(1) $A_1 \triangle A_2 = \mathbf{C}A_1 \triangle \mathbf{C}A_2$;

(2) $A_1 = A_1 \triangle A_2$ if and only if $A_2 = \varnothing$;

(3) For any sets A and A', there is a unique set E such that $E \triangle A = A'$, in fact, $E = A \triangle A'$;

(4) $\left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) \triangle \left(\bigcup_{\lambda\in\Lambda} A'_\lambda\right) \subseteq \bigcup_{\lambda\in\Lambda} (A_\lambda \triangle A'_\lambda)$.

这一页怎么读

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对应的完整证明：参见本讲义最后的「symdiff」证明专题。

2.36 第 36 页｜Basic Operation: Symmetric Difference

Basic Operation: Symmetric Difference

$$
\begin{aligned}
& A_1 \cap (A_2 \triangle A_3) = A_1 \cap [(A_2 \setminus A_3) \cup (A_3 \setminus A_2)] \\
= \;& [A_1 \cap (A_2 \setminus A_3)] \cup [A_1 \cap (A_3 \setminus A_2)] \\
= \;& \{A_1 \cap [A_2 \setminus (A_2 \cap A_3)]\} \cup \{A_1 \cap [A_3 \setminus (A_2 \cap A_3)]\} \\
= \;& [(A_1 \cap A_2) \setminus (A_1 \cap A_2 \cap A_3)] \cup [(A_1 \cap A_3) \setminus (A_1 \cap A_2 \cap A_3)] \\
= \;& [(A_1 \cap A_2) \cup (A_1 \cap A_3)] \setminus (A_1 \cap A_2 \cap A_3) \\
= \;& [A_1 \cap (A_2 \cup A_3)] \setminus (A_1 \cap A_2 \cap A_3)
\end{aligned}
$$

$$
\begin{aligned}
& (A_1 \cap A_2) \triangle (A_1 \cap A_3) = [(A_1 \cap A_2) \setminus (A_1 \cap A_3)] \cup [(A_1 \cap A_3) \setminus (A_1 \cap A_2)] \\
= \;& [(A_1 \cap A_2) \setminus (A_1 \cap A_2 \cap A_3)] \cup [(A_1 \cap A_3) \setminus (A_1 \cap A_2 \cap A_3)] \\
= \;& [(A_1 \cap A_2) \cup (A_1 \cap A_3)] \setminus (A_1 \cap A_2 \cap A_3) \\
= \;& [A_1 \cap (A_2 \cup A_3)] \setminus (A_1 \cap A_2 \cap A_3)
\end{aligned}
$$

这一页怎么读

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2.37 第 37 页｜Basic Operation: Symmetric Difference

Basic Operation: Symmetric Difference

Proof of (1).

$$\begin{aligned} \mathbf{C}A_1 \triangle \mathbf{C}A_2 &= (\mathbf{C}A_1 \setminus \mathbf{C}A_2) \cup (\mathbf{C}A_2 \setminus \mathbf{C}A_1) \\ &= (\mathbf{C}A_1 \cap A_2) \cup (\mathbf{C}A_2 \cap A_1) \\ &= (A_2 \setminus A_1) \cup (A_1 \setminus A_2) = A_1 \triangle A_2 \end{aligned}$$

Proof of (2). We first check $A_2 \subseteq A_1$. Otherwise, there exists $x_0 \in A_2 \setminus A_1$ and $A_1 \cup \{x_0\} \subseteq A_1 \triangle A_2 = A_1$, which is a contradiction!

If $A_2 \subseteq A_1$, then $A_1 = A_1 \triangle A_2 = A_1 \setminus A_2$ and hence $A_2 = \varnothing$.

Proof of (3).

$$E = E \triangle \varnothing = E \triangle (A \triangle A) = (E \triangle A) \triangle A = A' \triangle A.$$

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2.38 第 38 页｜Basic Operation: Symmetric Difference

Basic Operation: Symmetric Difference

Proof of (4).

$$
\begin{aligned}
& \left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) \triangle \left(\bigcup_{\lambda\in\Lambda} A'_\lambda\right) \\
= & \left[\left(\bigcup_{\lambda\in\Lambda} A_\lambda\right) \setminus \left(\bigcup_{\lambda\in\Lambda} A'_\lambda\right)\right] \cup \left[\left(\bigcup_{\lambda\in\Lambda} A'_\lambda\right) \setminus \left(\bigcup_{\lambda\in\Lambda} A_\lambda\right)\right] \\
\subseteq & \left(\bigcup_{\lambda\in\Lambda} \left[A_\lambda \setminus A'_\lambda\right]\right) \cup \left(\bigcup_{\lambda\in\Lambda} \left[A'_\lambda \setminus A_\lambda\right]\right) \\
= & \bigcup_{\lambda\in\Lambda} \left[A_\lambda \setminus A'_\lambda\right] \cup \left[A'_\lambda \setminus A_\lambda\right] \\
= & \bigcup_{\lambda\in\Lambda} \left(A_\lambda \triangle A'_\lambda\right).
\end{aligned}
$$

这一页怎么读

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对应的完整证明：参见本讲义最后的「symdiff」证明专题。

2.39 第 39 页｜Basic Operation: Symmetric DifferenceExample: Let A1, A

Basic Operation: Symmetric Difference

Example: Let A_1, A_2, and A_3 be three sets, then

(1) $(A_1 \cap A_2) \cup (A_2 \cap A_3) \cup (A_3 \cap A_1)$ represents a set formed by all elements belonging to at least two sets in A_1, A_2, and A_3;

(2) $(A_1 \cup A_2 \cup A_3) \setminus (A_1 \triangle A_2 \triangle A_3)$ indicates a set formed by all elements belonging to at least two sets but not three sets in A_1, A_2, and A_3;

(3) $(A_1 \triangle A_2 \triangle A_3) \setminus (A_1 \cap A_2 \cap A_3)$ indicates a set formed by all elements belonging to one set but not two sets in A_1, A_2, and A_3.

$$
\begin{aligned}
& (A_1 \cup A_2 \cup A_3) \setminus (A_1 \triangle A_2 \triangle A_3) \\
= \; & (A_1 \cap A_2) \cup (A_2 \cap A_3) \cup (A_3 \cap A_1) \setminus (A_1 \cap A_2 \cap A_3)
\end{aligned}
$$

$$
\begin{aligned}
& (A_1 \triangle A_2 \triangle A_3) \setminus (A_1 \cap A_2 \cap A_3) \\
= \; & [A_1 \setminus (A_2 \cup A_3)] \cup [A_2 \setminus (A_3 \cup A_1)] \cup [A_3 \setminus (A_1 \cup A_2)]
\end{aligned}
$$

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2.40 第 40 页｜Basic Operation: Symmetric DifferenceExample:

Basic Operation: Symmetric Difference

Example:

(1) Let A and A' be two subsets of the whole set S. If, for any $E \subseteq S$, $E \cap A = E \cup A'$, then $A = S$ and $A' = \varnothing$.

(2) Let A_1 and A_2 be two sets, then $A_1 = A_2$ if and only if there exists a set A_3 such that $A_1 \cap A_3 = A_2 \cap A_3$ and $A_1 \cup A_3 = A_2 \cup A_3$.

Proof. (1) If $E = S$, then $A = S$; Take $E = \mathbf{C}A$, then it is assumed that $\varnothing = \mathbf{C}A \cap A = \mathbf{C}A \cup A' \supseteq A'$, that is, $A' = \varnothing$.

(2) The necessity is obvious. Now we prove the sufficiency.

$$A_1 = A_1 \cup (A_3 \cap \mathbf{C}A_3) = (A_1 \cap A_3) \cup (A_1 \cap \mathbf{C}A_3)$$
$$A_2 = A_2 \cup (A_3 \cap \mathbf{C}A_3) = (A_2 \cap A_3) \cup (A_2 \cap \mathbf{C}A_3)$$

then $A_1 \cup A_3 = (A_1 \cap \mathbf{C}A_3) \cup A_3$ and $A_2 \cup A_3 = (A_2 \cap \mathbf{C}A_3) \cup A_3$. Note that $A_3 \cap (A_1 \cap \mathbf{C}A_3) = \varnothing = A_3 \cap (A_2 \cap \mathbf{C}A_3)$, then $A_1 \cap \mathbf{C}A_3 = A_2 \cap \mathbf{C}A_3$. Thus $A_1 = (A_1 \cap A_3) \cup (A_1 \cap \mathbf{C}A_3) = (A_2 \cap A_3) \cup (A_2 \cap \mathbf{C}A_3) = A_2$.

这一页怎么读

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2.41 第 41 页｜Basic Operation: Symmetric DifferenceQuestions. Prove:

Basic Operation: Symmetric Difference

Questions. Prove:

1. Let A, A' and E be three subsets of the whole set S, then $A' = \mathbf{C}(E \cap A) \cap (\mathbf{C}E \cup A)$ if and only if $\mathbf{C}A' = E$.
2. If $A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n \subseteq \cdots$, $A'_1 \subseteq A'_2 \subseteq \cdots \subseteq A'_n \subseteq \cdots$, then
$$\left(\bigcup_{n=1}^{\infty} A_n\right) \cap \left(\bigcup_{n=1}^{\infty} A'_n\right) = \bigcup_{n=1}^{\infty} (A_n \cap A'_n).$$
3. If $A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq \cdots$, $A'_1 \supseteq A'_2 \supseteq \cdots \supseteq A'_n \supseteq \cdots$, then
$$\left(\bigcap_{n=1}^{\infty} A_n\right) \cup \left(\bigcap_{n=1}^{\infty} A'_n\right) = \bigcap_{n=1}^{\infty} (A_n \cup A'_n).$$
4. Let A and A' and E and E' be four subsets of the whole set S.
   (1) If $A \cup A' = E \cup E'$, $A \cap E' = \varnothing$ and $A' \cap E = \varnothing$, then $A = E$ and $A' = E'$;
   (2) If $A \cup A' = E \cup E'$, let $A_1 = A \cap E$, $A_2 = A \cap E'$, then $A_1 \cup A_2 = A$.

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2.42 第 42 页｜Basic Operation: Symmetric Difference

Basic Operation: Symmetric Difference

Proof of 1.

$$\begin{aligned} A' &= \mathbf{C}(E\cap A)\cap(\mathbf{C}E\cup A) = (\mathbf{C}E\cup\mathbf{C}A)\cap(\mathbf{C}E\cup A)\\ &= \mathbf{C}E\cup(\mathbf{C}A\cap\cup A) = \mathbf{C}E\cup\varnothing = \mathbf{C}E. \end{aligned}$$

If $\mathbf{C}A' = E$, then

$$\begin{aligned} &\mathbf{C}(E\cap A)\cap(\mathbf{C}E\cup A) = \mathbf{C}\left(\mathbf{C}A'\cap A\right)\cap\left(A'\cup A\right)\\ =\ &\left(A'\cup\mathbf{C}A\right)\cap\left(A'\cup A\right) = A'\cup(\mathbf{C}A\cap A) = A'\cup\varnothing = A'. \end{aligned}$$

Conversely,

$$\begin{aligned} \mathbf{C}A' &= (E\cap A)\cup\mathbf{C}(\mathbf{C}E\cup A) = (E\cap A)\cup(E\cap\mathbf{C}A)\\ &= E\cap(A\cup\mathbf{C}A) = E\cap S = E. \end{aligned}$$

这一页怎么读

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2.43 第 43 页｜Basic Operation: Symmetric Difference

Basic Operation: Symmetric Difference

Proof of 4 (1).

$$A = A \cap (A \cup A') = A \cap (E \cup E') = (A \cap E) \cup (A \cap E') = A \cap E$$

implies $A \subseteq E$, and

$$A' = A' \cap (A \cup A') = A' \cap (E \cup E') = (A' \cap E) \cup (A' \cap E') = A' \cap E'$$

implies $A' \subseteq E'$, then $A \cup A' \subseteq E \cup E' = A \cup A'$, thus $A = E$ and $A' = E'$.

Proof of 4 (2).

$$A_1 \cup A_2 = (A \cap E) \cup (A \cap E') = A \cap (E \cup E') = A \cap (A \cup A') = A.$$

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2.44 第 44 页｜本页课件

Limit of Sequence of Sets

Similar to the limit of number sequence, which is a tool for infinite operation, it is also transplanted into set theory. We will see the important role of this operation in set representation. As we all know, the limit of monotone sequence can always be defined, which inspires us to consider the infinite operation of monotone set sequence.

Let $\{A_n; n \in \mathbb{N}\}$ be a set sequence. If

$$A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq \cdots$$

this set sequence is called a decreasing sequence of sets. In this case, we call its intersection $\bigcap_{n=1}^{\infty} A_n$ as the limit set of the set sequence $\{A_n; n \in \mathbb{N}\}$, denoted as $\lim_{n\to\infty} A_n$. If

$$A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n \subseteq \cdots$$

this set sequence is called an increasing sequence of sets. In this case, we call its union $\bigcup_{n=1}^{\infty} A_n$ as the limit set of the set sequence $\{A_n; n \in \mathbb{N}\}$, denoted as $\lim_{n\to\infty} A_n$.

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结合原文理解：先读“属于”的条件，再化成逻辑上的“且、或、非”。特别分清元素 x A 与子集 B A。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.45 第 45 页｜ Limit of Sequence of SetsExample: If An = [n, ∞) for n

Limit of Sequence of Sets

Example: If $A_n = [n, \infty)$ for $n = 1, 2, \cdots$, then $\lim_{n\to\infty} A_n = \varnothing$.

Example: Let $\{f_n; n \in \mathbb{N}\}$ be an increasing sequence of real-value functions on $\mathbb{R}$,

$$f_1(x) \leqslant f_2(x) \leqslant \cdots \leqslant f_n(x) \leqslant \cdots$$

and $\lim_{n\to\infty} f_n(x) = f(x)$. For any given real number t, define a sequence of sets

$$E_n = \{x \in \mathbb{R} \mid f_n(x) > t\}, \qquad n \in \mathbb{N}$$

then $E_1 \subseteq E_2 \subseteq \cdots \subseteq E_n \subseteq \cdots$, and

$$\lim_{n\to\infty} E_n = \bigcup_{n=1}^{\infty} \{x \in \mathbb{R} \mid f_n(x) > t\} = \{x \in \mathbb{R} \mid f(x) > t\},$$

that is,

$$\lim_{n\to\infty} \{x \in \mathbb{R} \mid f_n(x) > t\} = \{x \in \mathbb{R} \mid f(x) > t\}.$$

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2.46 第 46 页｜本页课件

Limit of Sequence of Sets

For general sequence of sets, the concept of upper and lower limit of sequence of sets can also be given in the same way as the upper and lower limits of sequence of numbers.

Let $\{A_n; n \in \mathbb{N}\}$ be a sequence of sets. Define

$$B_N = \bigcup_{n \geqslant N} A_n = \bigcup_{n=N}^{\infty} A_n, \qquad N \in \mathbb{N},$$

then $B_1 \supseteq B_2 \supseteq \cdots \supseteq B_n \supseteq \cdots$, $\{B_n; n \in \mathbb{N}\}$ is a decreasing sequence of sets.

$$\lim_{n\to\infty} B_n = \bigcap_{N=1}^{\infty} B_N = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n$$

is called the upper limit set of sequence of sets $\{A_n; n \in \mathbb{N}\}$, and is recorded as

$$\overline{\lim_{n\to\infty}} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n$$

这一页怎么读

概念主线：集合是对象的汇总。集合恒等式不依赖几何图形：证明两集合相等，最稳妥的方法是双向包含。

结合原文理解：先读“属于”的条件，再化成逻辑上的“且、或、非”。特别分清元素 x A 与子集 B A。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.47 第 47 页｜Limit of Sequence of SetsSimilarly, define

Limit of Sequence of Sets

Similarly, define

$$B_N = \bigcap_{n \geqslant N} A_n = \bigcap_{n=N}^{\infty} A_n, \qquad N \in \mathbb{N},$$

then $B_1 \subseteq B_2 \subseteq \cdots \subseteq B_n \subseteq \cdots$, $\{B_n; n \in \mathbb{N}\}$ is a increasing sequence of sets.

$$\lim_{n\to\infty} B_n = \bigcup_{N=1}^{\infty} B_N = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n$$

is called the lower limit set of sequence of sets $\{A_n; n \in \mathbb{N}\}$, and is recorded as

$$\underline{\lim_{n\to\infty}} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n$$

If the upper and lower limit sets of sequence of sets $\{A_n; n \in \mathbb{N}\}$ are equal, we call that its limit set exists and is equal to the upper and lower limit sets, $\lim_{n\to\infty} A_n$.

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2.48 第 48 页｜Limit of Sequence of Sets

Limit of Sequence of Sets

Theorem 5. Let $\{A_n; n \in \mathbb{N}\}$ be a sequence of sets, then

(1) $\overline{\lim_{n\to\infty}} A_n = \{x : \text{for any } N \in \mathbb{N}, \text{ there exists } n \geqslant N \text{ such that } x \in A_n\}$;

(2) $\underline{\lim_{n\to\infty}} A_n = \{x : \text{there exists } N \in \mathbb{N} \text{ such that, for any } n \geqslant N, \ x \in A_n\}$.

That is, the upper limit set of $\{A_n; n \in \mathbb{N}\}$ is formed by elements belonging to an infinite number of sets in $\{A_n; n \in \mathbb{N}\}$, the lower bound set of left $\{A_n; n \in \mathbb{N}\}$ is formed by elements belonging to all but finitely many sets in $\{A_n; n \in \mathbb{N}\}$. Therefore,

$$\overline{\lim_{n\to\infty}} A_n \supseteq \underline{\lim_{n\to\infty}} A_n.$$

这一页怎么读

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2.49 第 49 页｜Limit of Sequence of SetsQuestions. Prove:

Limit of Sequence of Sets

Questions. Prove:

1. Let $A_n = \left\{ \frac{m}{n} \mid m \in \mathbb{Z} \right\}$ and $B_n = \left(\frac{1}{n}, 1 + \frac{1}{n} \right)$ for $n \in \mathbb{N}$, then

$$\overline{\lim_{n\to\infty}} A_n = \mathbb{Q}, \qquad \underline{\lim_{n\to\infty}} A_n = \mathbb{Z}, \qquad \lim_{n\to\infty} B_n = (0, 1].$$

2. Let $\{f_n\}$ and f be real-valued functions defined on $\mathbb{R}$, and

$$\lim_{n\to\infty} f_n(x) = f(x), \qquad x \in \mathbb{R},$$

then, for $t \in \mathbb{R}$,

$$\{x \in \mathbb{R} \mid f(x) \leqslant t\} = \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{ x \in \mathbb{R} \mid f_n(x) < t + \frac{1}{k} \right\}.$$

3. If $a_n \to a$ $(n \to \infty)$, then

$$\bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left(a_n - \frac{1}{k}, a_n + \frac{1}{k} \right) = \{a\}.$$

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2.50 第 50 页｜Partition

Partition

A collection $\mathscr{X}$ of subsets of S forms a partition of S if they are pairwise disjoint (i.e., if any two different subsets belonging to $\mathscr{X}$ are disjoint) and if their union is S. That is, if each element of S belongs to one and only one of the subsets in $\mathscr{X}$. A partition of a set corresponds to the familiar idea of a classification of its elements.

A collection $\mathscr{X} = \{X_\gamma \mid \gamma \in \Gamma\}$ of subsets of S forms a partition of S if

(1) for any $\gamma \in \Gamma$, $\varnothing \neq X_\gamma \subset S$;

(2) $X_\gamma \cap X_{\gamma'} = \varnothing$ for $\gamma \neq \gamma'$;

(3) $S = \bigcup_{\gamma \in \Gamma} X_\gamma$.

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2.51 第 51 页｜Cartesian Product

Cartesian Product

Consider two sets S_1 and S_2; the set of pairs (x_1, x_2) where $x_1 \in S_1$ and $x_2 \in S_2$, is called their Cartesian product $S_1 \times S_2$. The order in which x_1, x_2 and S_1, S_2 are written in (x_1, x_2) and $S_1 \times S_2$ is essential.

Example: Let S_1 be the lower horizontal edge of a rectangle drawn in a plane and S_2 its left vertical edge. If x_1 is a point of S_1, x_2 a point of S_2, the pair (x_1, x_2) may be visualized as the intersection point of the vertical straight line through x_1 and the horizontal straight line through x_2. The product $S_1 \times S_2$ is then visualized as the region covered by the rectangle.

这一页怎么读

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读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.52 第 52 页｜Cartesian Product

Cartesian Product

More generally, consider N sets $S_1, \cdots, S_n, \cdots, S_N$. The set of N-tuples $(x_1, \cdots, x_n, \cdots, x_N)$, where $x_n \in S_n$, for every $n = 1, \ldots, N$ is the Cartesian product $S_1 \times \cdots \times S_n \times \cdots \times S_N$, also denoted $\prod_{n=1}^{N} S_n$. The order in which the x_n, and the S_n are written is again essential. The N-tuple $(x_1, \cdots, x_n, \cdots, x_N)$ is denoted by (x_n), and x_n, is called the nth coordinate (or the nth component) of (x_n).

Example: Consider a social system consisting of N agents. The nth agent must choose an action A_n, in a given set A_n of actions available to him. When each agent has made his choice, the outcome of the social activity is determined. Thus social activity is characterized by an N-tuple (a_n), an element of $\prod_{n=1}^{N} A_n$.

这一页怎么读

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2.53 第 53 页｜ Cartesian ProductTheorem 6. For family A = (Aλ | λ Λ)

Cartesian Product

Theorem 6. For family $\mathscr{A} = (A_\lambda \mid \lambda \in \Lambda)$ we have

$$A_\lambda \neq \varnothing \text{ for } \lambda \in \Lambda \iff \prod_{\lambda\in\Lambda} A_\lambda \neq \varnothing.$$

Theorem 7. If two families $\mathscr{A} = (A_\lambda \mid \lambda \in \Lambda)$ and $\mathscr{A}' = (A'_\lambda \mid \lambda \in \Lambda)$ have the same index set, then

$$\left\{ \begin{array}{c} A_\lambda \neq \varnothing \text{ for } \lambda \in \Lambda \\ \prod_{\lambda\in\Lambda} A_\lambda \subseteq \prod_{\lambda\in\Lambda} A'_\lambda \end{array} \right\} \iff \varnothing \subset A_\lambda \subseteq A'_\lambda \quad \text{for } \lambda \in \Lambda$$

and

$$\prod_{\lambda\in\Lambda} A_\lambda \cup \prod_{\lambda\in\Lambda} A'_\lambda \subseteq \prod_{\lambda\in\Lambda} (A_\lambda \cup A'_\lambda)$$
$$\prod_{\lambda\in\Lambda} A_\lambda \cap \prod_{\lambda\in\Lambda} A'_\lambda = \prod_{\lambda\in\Lambda} (A_\lambda \cap A'_\lambda).$$

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3 证明专题：考试应会的标准推导

专题的使用方法

这些专题沿着本课 PPT 的定理顺序补足主要推导；若原课件的命题另有附加假设，以该页原文为主，切勿将证明中的条件删掉。对于高维拓扑不动点定理，只陈述严格的适用条件并说明所需工具，不伪称给出了没有依据的完整高维证明。

3.1 专题：demorgan

证明｜De Morgan 定律的双向包含证明

证明 $(A\cup B)^c=A^c\cap B^c$。任取 x：$x\in(A\cup B)^c\Leftrightarrow x\notin A\cup B\Leftrightarrow(x\notin A\text{ 且 }x\notin B)\Leftrightarrow x\in A^c\cap B^c$。同理，$x\in(A\cap B)^c\Leftrightarrow(x\notin A\text{ 或 }x\notin B)\Leftrightarrow x\in A^c\cup B^c$。这里每一步都可以从右往左推，故为等价关系而非单向包含。

3.2 专题：symdiff

证明｜对称差的集合表达

定义 $A\triangle B=(A\setminus B)\cup(B\setminus A)$。因为 $x\in A\triangle B$ 等价于 $(x\in A\wedge x\notin B)$ 或 $(x\in B\wedge x\notin A)$，也就等价于 $x\in A\cup B$ 且 $x\notin A\cap B$。于是 $A\triangle B=(A\cup B)\setminus(A\cap B)$。交换 A,B 后条件不变，所以交换律成立。

3.3 专题：uncountable

证明｜Cantor 对角线法：$[0,1]$ 不可数

反设 $(0,1)$ 可以列为 $x_1,x_2,\ldots$。统一选用不以无穷多个 9 结尾的十进制展开。构造新小数 $y=0.y_1y_2\cdots$，其中若 x_n 的第 n 位是 1 就选 $y_n=2$，否则选 $y_n=1$。这样 $y\in(0,1)$，而且对每个 n，y 的第 n 位与 x_n 不同，故 $y\neq x_n$。这与列举已包含全部实数矛盾，所以 $(0,1)$ 不可数。

3.4 专题：image

证明｜像保持并集，但不一定保持交集

任取族 $A_i\subseteq S$。由定义 $y\in f(\bigcup_i A_i)\Leftrightarrow\exists x\in\bigcup_i A_i:f(x)=y\Leftrightarrow\exists i\,\exists x\in A_i:f(x)=y\Leftrightarrow y\in\bigcup_i f(A_i)$。对交集只有 $f(\bigcap_i A_i)\subseteq\bigcap_i f(A_i)$，因为右边的不同集合可能分别用不同的原像点表示同一个 y。反例：$f(1)=f(2)=0$，$A=\{1\}$，$B=\{2\}$，则 $f(A\cap B)=\varnothing$，而 $f(A)\cap f(B)=\{0\}$。对于逆像，$f^{-1}(\bigcap_i B_i)=\bigcap_i f^{-1}(B_i)$ 总成立。

4 自测清单

1. 你能不看课件写出本讲的核心定义及所有量词吗?
2. 你能逐条列出每一个定理的假设，并解释删去一条为何可能失败吗?
3. 你能在白纸上独立复述本讲证明专题中的每一个箭头吗?
4. 你能结合原课件的反例判断是否把充分条件错写成必要条件吗?